



2509 - Kernel-Phase Detection Limits for Planet Discovery with JWST

Cycle: 1, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

Filling the dearth of detections between directly-imaged and RV planets will test theories of planet formation across the full range of semi-major axes, connecting formation of close to wide separation gas giants and substellar companions. Direct detection of close-in companions is notoriously difficult: coronagraphs and PSF subtraction techniques are limited in separation and contrast. Non-redundant aperture masking interferometry can reveal companions well inside the PSF of a diffraction limited image, but the technique is severely flux-limited since the mask discards 95% of the gathered light. Kernel-phase interferometry applies similar interferometric techniques to an unobscured diffraction limited image, simulating the full telescope aperture as an interferometer composed of a grid of subapertures. We have developed a new faint companion detection pipeline using Bayesian model comparison of kernel-phases. We have demonstrated this pipeline on HST/NICMOS images of nearby brown dwarfs, refining astrometry of previously known companions and finding new companions. We have also characterized the detection limits, demonstrating significant sensitivity down to flux ratios of $\sim 10^{-2}$ at half L/D, reaching the planetary-mass regime for young targets. We now propose to apply modify our pipeline for use with JWST and apply it to PSF calibration targets observed through engineering, GTO, and DD-ERS programs. We will measure the detection limits of the kernel-phase analysis technique and determine best practice observing strategies in future cycles. Our pipeline will be made publicly available for straightforward kernel-phase analysis of imaging and AMI data.

OBSERVING DESCRIPTION

JWST Proposal 2509 (Created: Wednesday, March 31, 2021 at 1:01:36 AM Eastern Standard Time) - Overview

We propose to extend our current kernel-phase pipeline for use with JWST and measure on-sky sensitivity to close-in companions by analyzing PSF reference images observed with NIRCcam, MIRI, and NIRISS through Engineering, GTO, and DD-ERS programs. This dataset will allow us to characterize the sensitivity of kernel-phase analysis to faint close-in companions on real JWST images taken at a variety of wavelengths, using multiple cameras and observing strategies. Likely, the most important factor affecting measured sensitivity (beside image SNR) will be the observed wavelength and dither pattern since no JWST imaging cameras sample the PSF at greater than ~ 2.5 pix/FWHM with the exception of MIRI at long wavelengths. We may even be able to examine the effects of sun/roll angle and stability over time, with respect to the observation's proximity to mirror stacking events, on the measured sensitivity. Since kernel-phase is a generalization of closure phase analysis we can also apply our pipeline to NIRISS AMI mode observations to measure and compare sensitivity between analysis methods. Based on the measured detection limits, we will determine best practices for future observations utilizing kernel-phase analysis and make the pipeline publicly available for straightforward analysis of imaging and AMI data.